Stability-Based Early Stopping for Heterogeneous Effects in Sequential A/B Experiments

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Abstract

Sequential A/B experiments are commonly evaluated using aggregate treatment effects, although treatment effects may vary substantially across the population. When treatment effects are heterogeneous, an interim estimate of the conditional average treatment effect (CATE) can identify a subgroup that appears to experience meaningful harm even when the estimated harmful population is not yet stable.

This study investigates a stability-based stopping criterion for sequential A/B experiments with heterogeneous treatment effects. The proposed criterion requires both a sufficiently large estimated harmful population and persistence of that population across consecutive interim analyses. Harm is defined by an estimated CATE threshold, while population stability is quantified using Jaccard similarity between consecutive estimated harmful sets.

A simulation study with 500 replicated experiments compares two heterogeneous effect structures: a threshold-based effect and a smooth logistic effect. Each experiment contains 4,000 observations with equal randomized treatment allocation and is evaluated at six interim sample sizes. The simulation investigates the probability of stopping, agreement with the final decision, decision reversal, and stopping sample size.

The results show that requiring stability reduces early stopping and decision reversal in the threshold-based scenario, where the estimated harmful population is substantially unstable. At an illustrative Jaccard threshold of 0.75, stable stopping occurs in 31.4% of threshold-scenario replicates, compared with 39.2% under the naive criterion, while reversal decreases from 20.8% to 13.8%. In the smooth-effect scenario, stopping remains highly probable and reversal remains low. Increasing the Jaccard threshold produces a systematic trade-off between earlier stopping and decision stability.

These results suggest that the stability of the estimated affected population can provide useful information beyond the magnitude of an interim treatment effect when sequential decisions concern heterogeneous treatment effects.

1 Introduction

A/B experiments are often analyzed through an overall treatment effect, summarizing the difference in outcome rates between treatment and control groups. Such an analysis is appropriate when the treatment effect is approximately homogeneous. In many applications, however, the effect of an intervention can vary substantially across individuals or subpopulations.

The potential-outcomes framework represents this heterogeneity through individual-level treatment effects and their conditional averages [1]. Let $A_i \in \{0, 1\}$ denote treatment assignment and let $Y_i(1)$ and $Y_i(0)$ denote the corresponding potential outcomes. The individual treatment effect is

$$\tau_i = Y_i(1) - Y_i(0), \tag{1}$$

while the conditional average treatment effect is

$$\tau(x) = \mathbb{E}\left[Y(1) - Y(0) \mid X = x\right]. \tag{2}$$

The CATE is therefore a function of the observed covariates rather than a single population-level quantity. Estimation of heterogeneous treatment effects is consequently relevant when decisions depend on identifying subgroups for which treatment may be beneficial or harmful [2].

Sequential experimentation introduces an additional issue. Suppose that an experiment is evaluated at several interim sample sizes. The estimated CATE surface may change as new observations arrive. Consequently, the set of individuals or covariate values classified as experiencing meaningful harm may also change.

An early decision based only on the estimated size of this harmful population may therefore be sensitive to sampling variability. A harmful subgroup identified at one interim analysis may subsequently shrink, expand, or change composition.

This motivates a distinction between two questions:

1. Is the estimated harmful population sufficiently large to justify a stopping decision?
2. Is the estimated harmful population sufficiently stable across interim analyses that the decision is reproducible?

The present study investigates a simple extension of a CATE-based stopping criterion that explicitly incorporates the second question.

The central idea is to require persistence of the estimated harmful set across consecutive interim analyses. The stability of two estimated harmful sets is quantified using their Jaccard similarity. A stopping decision is permitted only when both the estimated harmful fraction exceeds a prespecified threshold and the similarity of consecutive harmful sets exceeds a stability threshold.

The objective is not to derive an optimal sequential testing boundary. Instead, the purpose of this simulation study is to investigate whether population-level stability of estimated heterogeneous effects provides useful additional information for sequential decision-making.

2 Statistical framework

2.1 Randomized A/B experiment

Consider a randomized experiment with observations

$$(X_i, A_i, Y_i), \qquad i = 1, \ldots, N, \tag{3}$$

where X_i is a baseline covariate, A_i is randomized treatment assignment, and Y_i is a binary outcome.

Treatment assignment is generated according to

$$A_i \sim \text{Bernoulli}(0.5). \tag{4}$$

The marginal average treatment effect is

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]. \tag{5}$$

Although the ATE is useful for describing the average effect, the present analysis focuses on treatment-effect heterogeneity.

2.2 CATE estimation

The conditional treatment effect is estimated from a logistic regression containing treatment, the baseline covariate, and their interaction:

$$\operatorname{logit}\{P(Y = 1 \mid A, X)\} = \beta_0 + \beta_1 A + \beta_2 X + \beta_3 AX. \tag{6}$$

For a given covariate value x, the fitted probabilities under treatment and control are

$$\widehat{p}_1(x) = P_{\widehat{\beta}}(Y = 1 \mid A = 1, X = x), \tag{7}$$

$$\widehat{p}_0(x) = P_{\widehat{\beta}}(Y = 1 \mid A = 0, X = x). \tag{8}$$

The estimated CATE is then defined on the probability scale as

$$\widehat{\tau}(x) = \widehat{p}_1(x) - \widehat{p}_0(x). \tag{9}$$

This counterfactual construction allows the estimated treatment effect to be evaluated for every observation in the interim analysis.

3 Definition of meaningful harm

The simulation defines meaningful harm using the estimated CATE threshold

$$\widehat{\tau}(X_i) \leq -0.02. \tag{10}$$

The corresponding estimated harmful set at interim analysis k is

$$\widehat{\mathcal{H}}_k = \{i : \widehat{\tau}_k(X_i) \leq -0.02\}. \tag{11}$$

The estimated harmful fraction is

$$\widehat{h}_k = \frac{|\widehat{\mathcal{H}}_k|}{n_k}, \tag{12}$$

where n_k denotes the sample size available at interim analysis k.

The basic CATE stopping criterion is therefore

$$\widehat{h}_k \geq h_0, \tag{13}$$

with

$$h_0 = 0.50. \tag{14}$$

Thus, under the naive rule, an experiment is considered eligible for stopping when at least 50% of the observations are classified as experiencing meaningful harm.

4 Stability-based stopping criterion

4.1 Jaccard similarity

To quantify the persistence of the estimated harmful population, consider two consecutive harmful sets,

$$\widehat{\mathcal{H}}_{k-1} \quad \text{and} \quad \widehat{\mathcal{H}}_k.$$

Their Jaccard similarity is

$$J_k = \frac{|\widehat{\mathcal{H}}_{k-1} \cap \widehat{\mathcal{H}}_k|}{|\widehat{\mathcal{H}}_{k-1} \cup \widehat{\mathcal{H}}_k|}. \tag{15}$$

The statistic satisfies

$$0 \leq J_k \leq 1, \tag{16}$$

with values closer to one indicating greater overlap between consecutive estimated harmful populations.

4.2 Stability-enhanced criterion

The stability-enhanced stopping rule requires

$$\widehat{h}_k \geq h_0 \tag{17}$$

and

$$J_k \geq J_0, \tag{18}$$

where J_0 is the prespecified Jaccard stability threshold.

The resulting rule can therefore be written as

$$\boxed{\widehat{h}_k \geq 0.50 \quad \text{and} \quad J_k \geq J_0.} \tag{19}$$

The analysis examines

$$J_0 \in \{0.50, 0.60, 0.70, 0.75, 0.80, 0.90\}. \tag{20}$$

The value $J_0 = 0.75$ is used as an illustrative operating point in the main results. It is not treated as an empirically optimized threshold.

5 Simulation design

5.1 General design

The simulation consists of 500 replicated experiments. Each experiment contains

$$N = 4000 \tag{21}$$

observations.

Treatment is assigned independently with probability 0.5. Six interim analyses are considered:

$$n_k \in \{1000, 1600, 2200, 2800, 3400, 4000\}. \tag{22}$$

The corresponding information fractions are

$$0.25,\ 0.40,\ 0.55,\ 0.70,\ 0.85,\ 1.00. \tag{23}$$

The baseline covariate is generated from a standard normal distribution,

$$X \sim N(0, 1). \tag{24}$$

Two heterogeneous treatment-effect scenarios are considered.

5.2 Scenario A: threshold heterogeneity

The first scenario contains a sharply defined subgroup for which treatment is harmful.

A subgroup is defined using the upper quartile of the baseline covariate:

$$\mathcal{H} = \{X > q_{0.75}\}. \tag{25}$$

The corresponding subgroup contains approximately 25% of the population.

Under control, the outcome probability is

$$p_0 = 0.10. \tag{26}$$

For observations outside the harmed subgroup, treatment has no effect,

$$p_1 - p_0 = 0. \tag{27}$$

Within the harmed subgroup,

$$p_1 - p_0 = -0.04. \tag{28}$$

Thus, the true individual-level treatment effect is

$$\tau(X) = \begin{cases} 0, & X \le q_{0.75}, \\ -0.04, & X > q_{0.75}. \end{cases} \tag{29}$$

The population-average true treatment effect is therefore approximately

$$\text{ATE} \approx -0.01. \tag{30}$$

This scenario represents a heterogeneous effect with a relatively sharp subgroup boundary.

5.3 Scenario B: smooth logistic heterogeneity

The second scenario replaces the sharp subgroup boundary with a smoothly varying treatment effect generated through a logistic functional form.

The treatment effect changes continuously with the baseline covariate rather than switching abruptly at a fixed threshold. Across the simulated population, the true CATE is approximately between -0.062 and -0.014, with an average effect of approximately

$$\text{ATE} \approx -0.037. \tag{31}$$

Unlike Scenario A, the smooth structure does not contain a sharply separated zero-effect population and harmed subgroup. Consequently, the estimated harmful population is expected to be more persistent across interim analyses.

The exact data-generating mechanism for this scenario is implemented in the accompanying R code.

6 Evaluation criteria

The simulation evaluates four primary characteristics of the stopping rules.

6.1 Stopping probability

For each scenario, the stopping probability is the proportion of replicated experiments for which a stopping criterion is satisfied at any interim analysis.

6.2 Agreement with the final decision

For each replicate, the final interim analysis at $n = 4000$ is treated as the reference decision. Agreement measures whether the sequential stopping decision is consistent with that final classification.

This is a simulation diagnostic rather than a claim that the final analysis constitutes a gold-standard clinical decision.

6.3 Decision reversal

A reversal occurs when an experiment stops early because the estimated harmful fraction satisfies the stopping criterion, but the final interim analysis no longer supports the corresponding harm classification.

This quantity directly measures the instability of early harm-based stopping.

6.4 Stopping sample size

The stopping sample size is the first interim sample size at which the relevant stopping criterion is satisfied.

The stability-enhanced rule is expected to require more information when the estimated harmful population is unstable.

7 Results

7.1 Evolution of the estimated harmful population

Figure 1 shows the mean estimated fraction classified as experiencing meaningful harm across the six interim analyses.

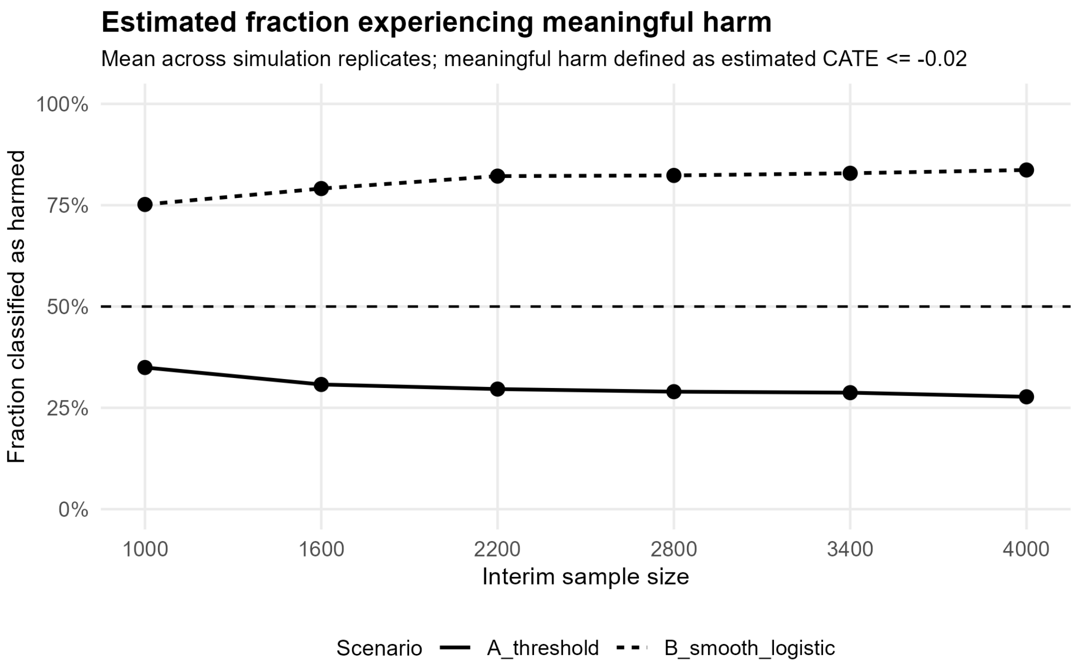

Figure 1: Mean estimated fraction of observations classified as experiencing meaningful harm across simulation replicates. Meaningful harm is defined as an estimated CATE less than or equal to -0.02. The horizontal dashed line represents the 50% stopping threshold.

In Scenario A, the estimated harmful fraction decreases from approximately 35% at the first interim analysis to approximately 28% at the final analysis.

In Scenario B, the estimated harmful fraction remains substantially larger, increasing from approximately 75% to approximately 84%. This difference between the two scenarios provides a useful contrast between an unstable threshold-type effect and a smoother heterogeneous effect.

7.2 Stability of the estimated harmful population

Figure 2 shows the Jaccard similarity between consecutive estimated harmful populations.

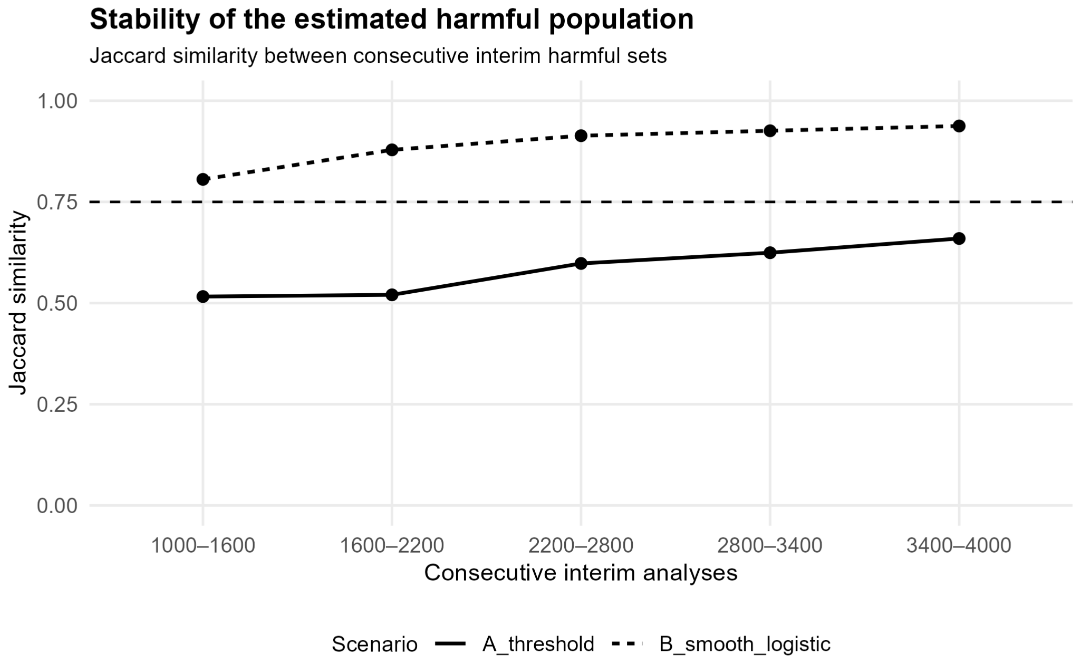

Figure 2: Jaccard similarity between consecutive estimated harmful populations. The dashed horizontal line indicates the illustrative stability threshold $J_0 = 0.75$.

Scenario A exhibits considerably lower stability. The Jaccard similarity starts at approximately 0.52 and increases to approximately 0.66 by the final transition. None of the observed mean similarities reaches the illustrative threshold of 0.75.

In contrast, Scenario B has substantially higher Jaccard similarity. The similarity increases from approximately 0.81 at the first transition to approximately 0.94 at the final transition.

The result indicates that the two scenarios differ not only in the magnitude of their estimated harmful fractions but also in the persistence of the estimated harmful population.

7.3 Sensitivity to the stability threshold

Figure 3 shows the probability of stopping under different Jaccard thresholds.

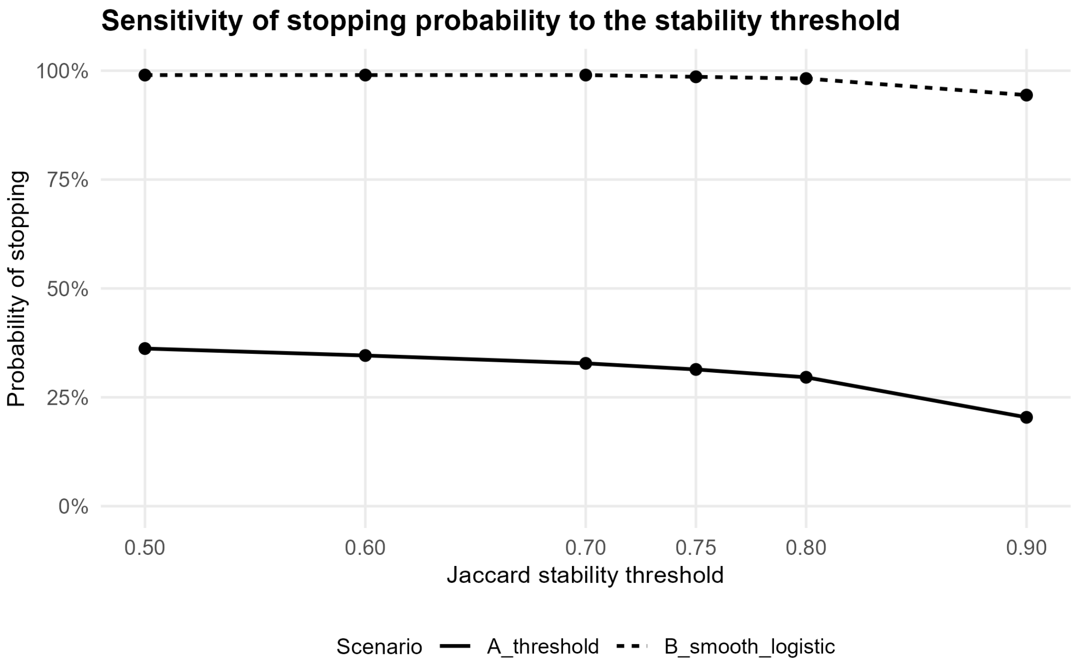

Figure 3: Probability of stopping under the stability-enhanced criterion as a function of the Jaccard stability threshold.

For Scenario A, increasing the Jaccard threshold progressively reduces the probability of stopping. It decreases from 36.2% at $J_0 = 0.50$ to 20.4% at $J_0 = 0.90$.

Scenario B is substantially less sensitive. The stopping probability remains 99.0% at thresholds through 0.70 and decreases to 94.4% at $J_0 = 0.90$.

This pattern illustrates the intended role of the stability criterion: increasing the requirement for persistence primarily suppresses stopping in the scenario in which the estimated harmful population is unstable.

7.4 Decision reversal

Figure 4 presents the probability of decision reversal.

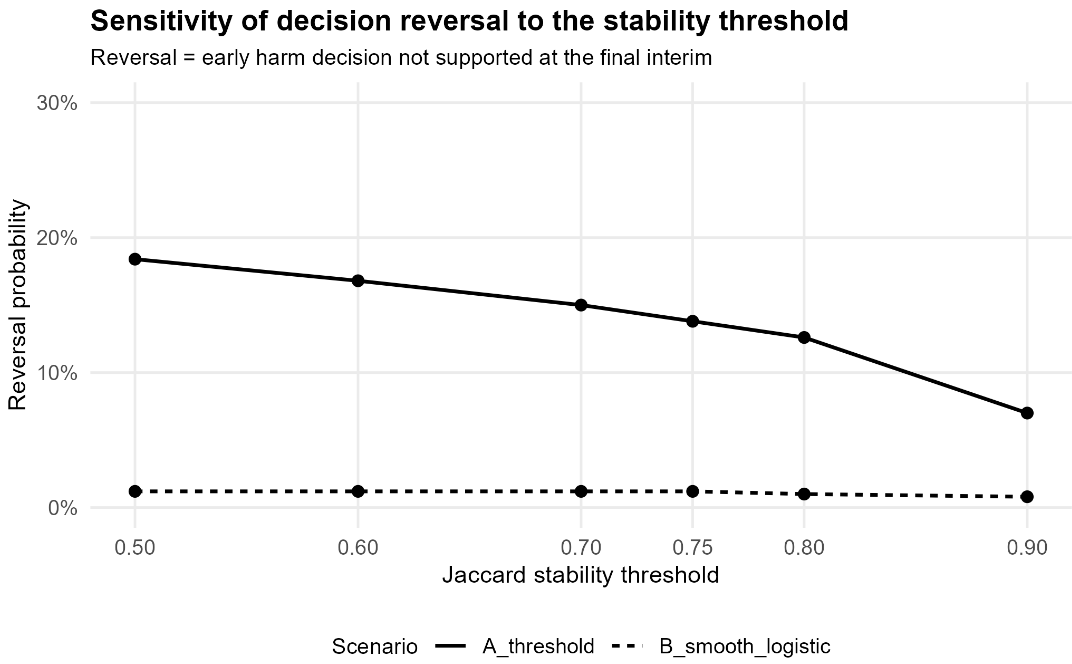

Figure 4: Probability of decision reversal as a function of the Jaccard stability threshold. A reversal occurs when an early stopping decision based on estimated harm is not supported by the final interim analysis.

The reversal probability decreases systematically for Scenario A as the stability threshold increases. It falls from 18.4% at $J_0 = 0.50$ to 7.0% at $J_0 = 0.90$.

The corresponding probability remains close to 1% in Scenario B across the range of thresholds.

Thus, the principal benefit of increasing the stability requirement in the threshold scenario is a reduction in early decisions that subsequently fail to persist.

7.5 Stopping-time trade-off

The reduction in early stopping is accompanied by an increase in the amount of information required before stopping.

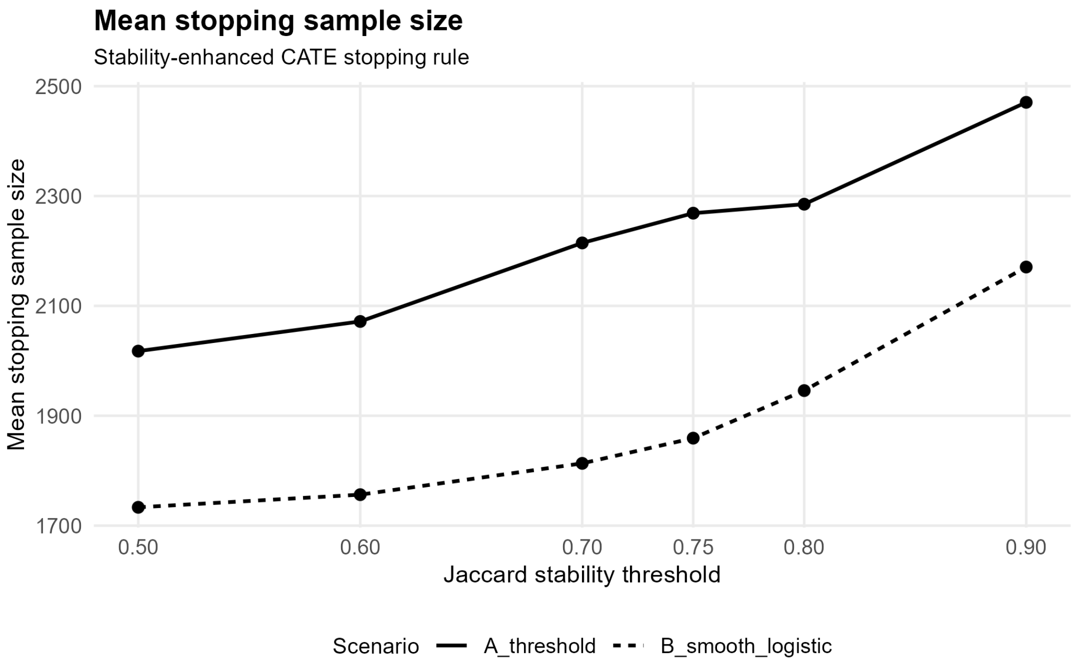

Figure 5: Mean stopping sample size under the stability-enhanced stopping rule as a function of the Jaccard threshold.

For Scenario A, the mean stable stopping sample size increases from approximately 2,018 at $J_0 = 0.50$ to approximately 2,471 at $J_0 = 0.90$.

For Scenario B, the corresponding increase is from approximately 1,733 to approximately 2,171.

The stability requirement therefore introduces an explicit efficiency versus stability trade-off: more stringent persistence requirements reduce early reversal but delay stopping.

7.6 Illustrative operating point: $J_0 = 0.75$

For a more focused comparison, Table 1 summarizes the results at the illustrative Jaccard threshold

$$J_0 = 0.75. \tag{32}$$

Table 1: Stopping performance at the illustrative stability threshold $J_0 = 0.75$.

	Scenario A	Scenario B
Naive stopping probability	39.2%	99.0%
Stable stopping probability	31.4%	98.6%
Naive agreement with final decision	79.2%	98.8%
Stable agreement with final decision	85.4%	98.4%
Naive reversal probability	20.8%	1.2%
Stable reversal probability	13.8%	1.2%
Naive mean stopping sample size	1,898	1,677
Stable mean stopping sample size	2,269	1,859

For Scenario A, introducing the stability criterion at $J_0 = 0.75$ reduces the stopping probability from 39.2% to 31.4%. At the same time, agreement with the final decision increases from 79.2% to 85.4%, while reversal decreases from 20.8% to 13.8%.

The corresponding change in Scenario B is considerably smaller. The stopping probability remains 98.6%, and the reversal probability remains 1.2%.

The stable rule therefore changes the sequential behavior most strongly in the scenario characterized by lower harmful-set stability.

8 Discussion

The simulation demonstrates a simple but important distinction between estimating the magnitude of heterogeneous treatment effects and assessing the stability of the population identified as being affected.

In Scenario A, the estimated harmful population is unstable across interim analyses. The corresponding Jaccard similarities are substantially below the illustrative threshold of 0.75. A stopping rule based only on the harmful fraction can consequently stop at an interim analysis even when the estimated harmful classification is not persistent.

Requiring a minimum degree of population stability makes the stopping rule more conservative in this setting. The principal effect is not merely to reduce the number of stopping decisions. It is to reduce decisions that do not remain supported at the final interim analysis.

Scenario B provides a contrasting case. The estimated harmful population is much more stable, with Jaccard similarities consistently above approximately 0.80. Consequently, the stability requirement has relatively little effect on the stopping probability.

This behavior is desirable for the proposed criterion. If the estimated affected population is already persistent, requiring stability should not substantially obstruct stopping. Conversely, when the population is unstable, the additional criterion should delay or prevent early decisions.

The sensitivity analysis also demonstrates that the stability threshold controls a meaningful operating characteristic. Higher values of J_0 reduce reversal probability but increase the average stopping sample size. There is therefore no universally optimal threshold based on the present simulation alone.

The threshold $J_0 = 0.75$ is consequently used only as an illustrative operating point. The sensitivity analysis is more important than the particular choice of 0.75 because it shows the qualitative relationship between stability requirements, reversal, and information requirements.

9 Limitations

Several limitations should be emphasized.

First, the simulation uses a single baseline covariate and a relatively simple logistic CATE estimation model. Real A/B experiments may contain many covariates, nonlinear interactions, missing data, or model misspecification.

Second, the harmfulness threshold of -0.02 and harmful-fraction threshold of 50% are simulation parameters rather than universally meaningful decision boundaries. Their interpretation depends on the application.

Third, the Jaccard threshold is not selected through an optimality criterion. The values considered in the sensitivity analysis are intended to demonstrate the operating characteristics of the proposed rule rather than establish a universally appropriate threshold.

Fourth, the simulation evaluates agreement with the final interim analysis rather than an independent external truth criterion. The final decision is therefore a reference point for assessing sequential instability, not a formal gold standard.

Finally, the CATE estimates are obtained from a parametric logistic model. The behavior of the proposed stability criterion under more flexible machine-learning estimators remains an important area for future work.

10 Conclusion

This simulation study examined a stability-based extension of CATE-driven early stopping in sequential A/B experiments.

The proposed rule combines two conditions:

$$\widehat{h}_k \geq h_0 \tag{33}$$

and

$$J_k \geq J_0. \tag{34}$$

The first condition asks whether a sufficiently large fraction of the population is estimated to experience meaningful harm. The second asks whether the estimated harmful population is persistent across consecutive interim analyses.

The simulations show that these two requirements behave differently under different forms of treatment-effect heterogeneity. In the threshold scenario, where the estimated harmful population is substantially less stable, adding the stability criterion reduces early stopping and decision reversal. In the smooth logistic scenario, where the estimated harmful population is more persistent, the additional criterion has a much smaller effect.

At the illustrative threshold $J_0 = 0.75$, the stability-enhanced rule reduces reversal from 20.8% to 13.8% in the threshold scenario, while retaining a high stopping probability of 98.6% in the smooth logistic scenario.

The main implication is therefore not that a particular Jaccard threshold is universally optimal. Rather, the simulation suggests that the stability of the estimated affected population can be treated as an additional diagnostic for sequential decisions involving heterogeneous treatment effects.

References

[1] Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701.

[2] Athey, S., & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27), 7353–7360.

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A Reproducibility

All simulation experiments and analyses are implemented in R.

The analysis is organized into sequential scripts covering:

1. data generation,
2. CATE estimation,
3. validation,
4. repeated sequential simulation,
5. stopping-rule evaluation,
6. stability-based stopping,
7. CATE stability metrics,
8. stability-threshold sensitivity analysis, and
9. final figure generation.

The accompanying repository contains the R scripts and generated results needed to reproduce the figures and numerical summaries presented in this paper.

The simulation uses a fixed random seed for the initial data-generation experiment, while the repeated simulation framework controls the generation of replicated experiments.